

Mathematics and Welfare: Optimal Investment Pre and Post LLMs

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Abstract

Pure mathematics, and applied mathematics pursued without a client, produce non-rival ideas whose social value depends on how many uses they reach, what each use costs to make, and how long the match takes. Large language models move all three at once. We write the value of an unmotivated result as a function of reach, adoption cost, and lag, derive the elasticity of optimal research spending with respect to that value, and compute the implied shift in the ratio of investment in mathematics to investment in the professions that train rival human capital. The channel sizes are assumptions. Under conservative ones the planner's ratio rises by nearly an order of magnitude, under central ones by two, and the shift is largest for the research whose old matching mechanism worked worst.

1 Six students

The pure mathematics class I graduated with at the University of New South Wales numbered about half a dozen. Down the hill, law and commerce enrolled students by the thousand; today the law faculty has about 2,600 students ([UNSW, 2021](#)) and the business school about 24,000 ([UNSW, 2024](#)). Nobody found this odd. The ratio looked like a market outcome: the professions paid, and pure mathematics, on the whole, did not.

The disparity is not peculiar to one campus. In the United States, departments awarded about 1,450 doctorates in mathematics and applied mathematics in the 2021–22 academic year ([American Mathematical Society, 2023](#)), against roughly 30,000 law degrees ([American Bar Association, 2024](#)) and roughly 200,000 master's degrees in business ([National Center for Education Statistics, 2023](#)). On the research side, the National Science Foundation's Division of Mathematical Sciences requested \$248 million for fiscal year 2025 ([National Science Foundation, 2024](#)), about one part in a hundred thousand of national output.

This paper asks what that ratio should be, and specifically how it should change now that large language models exist. We take the pre-LLM allocation as a benchmark, write down how a model changes the value of an unmotivated mathematical result, and derive the posterior. The argument is theoretical. The empirical anchors are the estimates that already exist for the return to research in general, and the observed lag between the founding of a mathematical field and its first external application.

2 What counts as unmotivated

By *unmotivated* mathematics we mean research whose direction is chosen by criteria internal to mathematics: generality, difficulty, elegance, the wish to see what happens. Pure mathematics is

the obvious case. So is a large share of applied mathematics, where the applied label describes the vocabulary rather than the presence of a customer. Radon had no client in 1917 for the transform that now runs every CT scanner; Sinkhorn was proving a theorem about doubly stochastic matrices, not balancing betting pools.

The defining property is that the eventual user is not known at the time of the work. Motivated research knows its user and is designed for that user, so its value is roughly the value of that one application. Unmotivated research has, at the time it is done, a distribution of possible users, most of whom will never hear of it. Its value is an expectation over that distribution, and the expectation is governed by how efficiently results travel.

The historical record on travel is not encouraging. A survey of the pure-mathematics fields in the 1959 Mathematics Subject Classification finds a median gap of 68 years between a field's founding and its first documented external application, with quartiles at 37 and 107 years (Cotton, 2026a). Hardy's number theory took a little under a century to become public-key cryptography. Kepler's sphere packing took three centuries to become lattice cryptography. Part of each lag is waiting for a problem, or a complementary technology, to exist at all. The rest is transmission, which until recently meant a person who happened to know both the problem and the theorem. The case studies in Section 4 show that part was large.

3 Prior estimates of value

Three literatures bear on the absolute question, and all three say the number is large.

Returns to research in general. Jones and Summers (2020) compute the economy-wide social return to innovation investment and find a benefit-cost ratio above ten under central parameters, at least four under conservative ones, and above twenty once international spillovers and health benefits are counted. The estimate is for research and development as a whole, not mathematics, but mathematics sits at the upstream end of the chain, where the spillovers are largest.

Gross attribution to the mathematical sciences. Deloitte (2013) attribute 2.8 million UK jobs and £208 billion of gross value added, about sixteen per cent of the total, to occupations that use tools derived from mathematical-science research. The Academy for the Mathematical Sciences updated the figure to £495 billion for 2023 (Academy for the Mathematical Sciences, 2024). These are gross measures with an occupational attribution rule, not marginal returns to another pound of research, and they should be read as an upper envelope rather than an estimate of the derivative.

The underinvestment argument. Nelson (1959) and Arrow (1962) argued that private markets underinvest in basic research because the returns cannot be appropriated by the investor. Romer (1990) made the mechanism explicit: ideas are non-rival, so their social value scales with the number of users while the private value does not. The argument is older than large language models and does not depend on them. It implies that the pre-LLM level of spending was a private optimum sitting below the social one.

We proceed with that benchmark anyway, because the question of interest is the direction and size of the shift, and the shift can be computed without settling the level.

4 The value of an unmotivated result

Fix a result r produced by unmotivated research. Index its potential uses by $i = 1, \dots, K$. Use i has gross value $v_i > 0$ if the result is applied to it, an adoption cost $c_i > 0$ covering the work of finding, understanding, and implementing the result in that context, and, if the match is ever made, a lag $T_i \geq 0$ before it happens. Whether the match is made at all is the encounter indicator $A_i \in \{0, 1\}$. The result is non-rival: use i does not diminish use j . With social discount rate ρ , the realized social value is

$$V(r) = \sum_{i=1}^K A_i (v_i - c_i)^+ e^{-\rho T_i}. \quad (1)$$

In English: add up every use the result ever reaches, net of what each use costs to adopt, and discount each by how long it took. Four primitives enter: the number of potential uses K , the encounter indicators A_i , the adoption costs c_i , and the lags T_i . Everything that follows is about how each moves.

To get closed forms, take the v_i as independent draws from a Pareto distribution with minimum v_0 and tail index $\alpha > 1$, so that $\Pr(v > x) = (x/v_0)^{-\alpha}$ for $x \geq v_0$. Heavy tails are the empirically correct assumption for the value of innovations: a small fraction of patents carries most of the value (Scherer and Harhoff, 2000). Take a common adoption cost $c \geq v_0$, a common encounter probability $q = \Pr(A_i = 1)$, and a common conditional expected discount factor $D = \mathbb{E}[e^{-\rho T_i} \mid A_i = 1]$, with the value v_i independent of the pair (A_i, T_i) . Then

$$g(c) := \mathbb{E}[(v - c)^+] = \frac{v_0^\alpha c^{1-\alpha}}{\alpha - 1}, \quad \mathbb{E} V = K \cdot q \cdot g(c) \cdot D. \quad (2)$$

The discount factor is an expectation of $e^{-\rho T}$, not $e^{-\rho}$ raised to the expected lag; by Jensen's inequality the two differ, and the first is the larger. It is convenient to write $D = e^{-\rho \tilde{\tau}}$ with $\tilde{\tau} = -\rho^{-1} \log D$ the *certainty-equivalent lag*, which is what the lag figures below refer to. The elasticities are immediate. Value is linear in the encounter probability, has elasticity $-(\alpha - 1)$ in adoption cost on the domain $c \geq v_0$, and falls at rate ρ per year of certainty-equivalent lag.

Definition 1 (Reach). The *reach* of a result is Kq , the expected number of uses at which the result is encountered at all, whether or not it is then adopted.

Two assumptions sit inside (2). The value of a use is independent of whether and when it is reached, so the uses a result reaches are drawn from the same distribution as the uses it misses; a use valuable enough to hire a mathematician was more likely to be reached under the old regime, and to that extent the reach ratio below is generous. And reach and lag are two views of one matching process: reach is ever matched within a fixed horizon, D is the expected discount conditional on matching, and with those definitions the product form is exact.

Before large language models, a use could encounter a result only if a person at that use was *bilingual*: fluent in the problem and fluent enough in the mathematics to recognize the match. The encounter probability of an unmotivated result was the probability that such a person was present. This is the bottleneck documented case by case in Cotton (2026b): the same invariant was defined four times under four names in four literatures because no one at any of the three later sites had encountered the first.

5 What a language model changes

A large language model is, for present purposes, a device that makes every result in its training corpus or retrieval index available at every use simultaneously. Three consequences follow, one per primitive.

Reach. A result that enters the corpus is present in every copy of the model. The person at the use no longer needs to know the theorem. They need to describe the problem, and the model supplies the match. The encounter probability rises from the chance that a bilingual person was present to the chance that a model is consulted, which is approaching one. If bilingual people were present at one use in ten, q rises by a factor of ten; if at one in thirty, thirty. Write $n = q_1/q_0$ for this ratio, with subscripts 0 and 1 for the regimes before and after.

Adoption cost. Once the match is made, the model can also write the implementation. The cost of taking a theorem from the page to a working change in an application falls from weeks of a specialist’s time to minutes of anyone’s. By (2) the effect on value is $m^{\alpha-1}$ for a cost reduction by factor m , provided the reduced cost c/m still exceeds v_0 ; below that floor every use is adopted and the multiplier is smaller. This channel is weaker than it sounds when tails are heavy: with α near one, value is dominated by the few enormous uses that would have paid the old cost anyway. It is a factor of a few, not tens.

Lag. A result posted in the morning can be retrieved and read by a model in the afternoon. The lag from publication to availability at every use collapses from the decades the historical record shows to the same day. What remains is the wait for a problem the result fits to arise, and that is a property of the problem stream, not of transmission. The lag multiplier is $D_1/D_0 = e^{\rho(\tilde{\tau}_0 - \tilde{\tau}_1)}$. At a three per cent discount rate, removing forty years of certainty-equivalent lag multiplies present value by $e^{0.03 \times 40} \approx 3.3$. At five per cent, removing 68 years gives about thirty.

The 68-year median is a field-level figure whose clock starts at the field’s founding; the lag T_i in (1) starts when a particular result is published, and the two are different quantities. What the record establishes is that lags of decades were normal. The year counts used below are assumptions on that scale.

Collecting the three channels, the ratio of post to pre expected value is

$$\kappa = n \cdot m^{\alpha-1} \cdot \frac{D_1}{D_0} = n \cdot m^{\alpha-1} \cdot e^{\rho(\tilde{\tau}_0 - \tilde{\tau}_1)}. \quad (3)$$

Proposition 1 (The multiplier is largest where the old matching was worst). *Hold m , α , ρ , q_1 , and D_1 fixed. Then κ is decreasing in the pre-LLM encounter probability q_0 and decreasing in the pre-LLM discount factor D_0 , that is, increasing in the pre-LLM certainty-equivalent lag $\tilde{\tau}_0$. For research with a single designed-in user who possessed it at the outset, $q_0 = q_1 = 1$ and $D_0 = D_1$, so $\kappa = m^{\alpha-1}$: only the adoption-cost channel remains.*

Proof. Immediate from (3), since $n = q_1/q_0$ and the lag factor is D_1/D_0 . □

Targeted research already knew its customer, and the model helps it at the margin. Untargeted research did not, and the model supplies the one thing it lacked: a matching mechanism. The gain from the mechanism is proportional to how badly the old one worked, and the historical lags say it worked badly.

6 Optimal investment

Let a planner spend S dollars on unmotivated mathematics and obtain $I(S) = aS^\gamma$ results, $0 < \gamma < 1$, each of expected value V . The planner maximizes $aVS^\gamma - S$. The first-order condition $\gamma aVS^{\gamma-1} = 1$ gives

$$S^* = (\gamma aV)^{1/(1-\gamma)}, \quad \frac{\partial \log S^*}{\partial \log V} = \frac{\partial \log S^*}{\partial \log a} = \frac{1}{1-\gamma} \geq 1. \quad (4)$$

Proposition 2 (Spending elasticity). *If the expected value of a result is multiplied by κ , optimal spending is multiplied by $\kappa^{1/(1-\gamma)}$. The elasticity is at least one, and it is larger the weaker the diminishing returns: as $\gamma \rightarrow 0$ it tends to one, and at $\gamma = \frac{1}{2}$ it equals two.*

Why does the elasticity exceed one? With diminishing returns the marginal result is cheap when spending is low and expensive when spending is high. If each result becomes ten times more valuable, the planner keeps buying until the marginal result costs ten times more, and under a square-root ideas function that means spending one hundred times more. The conservative reading is the limit $\gamma \rightarrow 0$, where results barely respond to spending and the planner's response is proportional to value. We report both ends.

The curvature γ is static, the returns to more spending at a point in time. It is a different object from the fall in research productivity over calendar time that [Bloom et al. \(2020\)](#) document, which is a statement about a .

7 The ratio to law and commerce

Now let the planner split a fixed budget B between mathematics, M , and a professional sector, L , that trains rival human capital: a lawyer's hour is used once. With the same functional form in each sector and a common γ , the problem is

$$\max_{M+L=B} a_M V_M M^\gamma + a_L V_L L^\gamma, \quad \text{giving} \quad \frac{M^*}{L^*} = \left(\frac{a_M V_M}{a_L V_L} \right)^{1/(1-\gamma)}. \quad (5)$$

This is the same elasticity as (4), so the two sections agree. If a language model multiplies V_M by κ_M and V_L by κ_L , the ratio is multiplied by $(\kappa_M/\kappa_L)^{1/(1-\gamma)}$.

The two multipliers may move in opposite directions. The product of mathematics is a non-rival idea, and the model raises its reach. The product of a law or commerce degree is a person's capacity to perform tasks, and the model performs a share of those tasks itself. [Eloundou et al. \(2023\)](#) rate legal services among the occupations most exposed to language models. Exposure is not substitution, so we take $\kappa_L = 1$ as the baseline and $\kappa_L < 1$ as the sensitivity.

Remark 1 (Training users versus training producers). The argument concerns spending on the production of unmotivated mathematics, including the pipeline that produces the people who produce it. It does not imply that every applied practitioner should be trained in more mathematics. The opposite is closer to the truth: the model is now the bilingual person in the room, so the applied practitioner needs less mathematics than before, not more. The mathematics still has to be produced by someone, at a spend that now looks far too small.

8 The posterior

The posterior is the benchmark ratio updated by the multipliers in (5). Table 1 sets out a low, central, and high value for each channel and reads off the result. The reach multiplier n is the ratio of the post-LLM to the pre-LLM encounter probability; one worker in ten is the share Deloitte (2013) count as mathematically occupied, and one problem in thirty is a guess at how often such a worker was actually in the room. The adoption-cost multiplier uses $m = 3, 10$, and 10 with tail indices 1.2, 1.5, and 2, on the domain $c \geq mv_0$. The lag multiplier uses discount rates of three and five per cent and reductions in certainty-equivalent lag of twenty-five, forty, and sixty-eight years. Channel entries are rounded before multiplying.

Channel	Low	Central	High
Reach n	3	10	30
Adoption cost $m^{\alpha-1}$	1.2	3	10
Lag D_1/D_0	2.1	3.3	30
Value multiplier κ_M	7.6	99	9000
Professional multiplier κ_L	1.0	1.0	0.7
Ratio multiplier, $\gamma \rightarrow 0$	7.6	99	13000
Ratio multiplier, $\gamma = \frac{1}{2}$	58	9800	1.7×10^8

Table 1: Channel multipliers and the implied change in the ratio of spending on unmotivated mathematics to spending on the professions, from (5). The elasticity is $1/(1 - \gamma)$: one in the limit $\gamma \rightarrow 0$, two at $\gamma = \frac{1}{2}$.

The high column is not a forecast. It is there to show the range once the channels compound. The low column, with a reach multiplier of three, a nearly useless cost channel, and a modest lag improvement, moves the ratio by a factor of nearly eight. The central column moves it by two orders of magnitude at the conservative elasticity, and by four at $\gamma = \frac{1}{2}$. A reach multiplier near one would mean a bilingual person was already present at most uses, and the case studies say otherwise.

Scaling the American degree counts of Section 1 by the same multipliers, with a degree in each sector as a unit of investment, the pre-LLM ratio of mathematics doctorates to law and business degrees combined was about one to 160. Under the low column it becomes one to 20; under the central column, at the conservative elasticity, about one to two. On a campus, six students against several thousand becomes fifty under the low column and a lecture theatre under the central one. Degree counts are proxies for spending, not measures of it.

9 Objections

The ideas were already free. Journals were open, or nearly so, for decades before language models, and the lags were still measured in generations. Free to read is not free to find, and not free to recognize as relevant. The scarce input was never the paper. It was the person who had read the paper and was also standing next to the problem, and that person is what the model replaces.

Ideas are getting harder to find. Bloom et al. (2020) document falling research productivity across fields. This is a statement about a in (4), the production side. The present argument is about V , the value side, and the two enter multiplicatively. A field can be harder to advance and, at the same time, far more valuable to advance.

The model will do the mathematics too. Perhaps. If it does, a rises, and by (4) optimal spending rises with it, by the same elasticity as for a rise in V . The question this paper answers is how much unmotivated mathematics society should want, and a cheaper production technology increases both the quantity and the optimal spend. Who or what produces it is a separate matter, and the human share of the pipeline is a question about labour markets rather than about welfare.

The benchmark was not an equilibrium. Agreed, and Section 3 says so. The Nelson–Arrow argument says the pre-LLM level of spending on mathematics already sat below the social optimum, so the posterior multiplies a number that was too small to begin with.

Hardy. Hardy (1940) wrote that number theory would never be useful, and was wrong within his readers’ lifetimes. He was wrong about the destination, but he was not wrong about the mechanism as it stood in 1940: the results had nowhere to go but the heads of other number theorists, and the odds that one of them would be in the room when the cryptographers needed a trapdoor were poor. The odds are no longer poor. They are, approximately, one.

10 The misalignment declaration

Twenty-five Fields medalists have signed a declaration, *A Severe Misalignment of AI in Mathematics* (Avila et al., 2026), arguing that the push to solve mathematical problems as a benchmark is detrimental to the science. Their central distinction is between solving problems and understanding: solving is only a tool and proxy for conceptual understanding, and the mass production of true/false statements could destroy fertile ground instead of breathing life into new ideas. They also fear the loss of the human transmission chain, the long process of talks, discussions, and simplifications that turns a difficult proof into a textbook chapter. The declaration allows, in one sentence, that AI offers the potential of enhancing and accelerating genuine mathematical study and understanding. What follows is about that sentence.

The model above offers a brighter reading of precisely that distinction. If understanding is the goal, cheaper access to solutions is a gift. A solved problem remains available to understand, simplify, generalise, connect to other fields, and teach. Knowing that a route exists makes exploring the landscape considerably more productive.

Consider the research directions currently abandoned because investigating them would consume six months, or the connections never pursued because acquiring the background would take two years. Lower those costs and a mathematician can entertain more conjectures, investigate more examples, and venture further beyond their specialism. In the notation of (4) this raises a , the number of results per unit of spend, and it does so alongside the rise in V that is the subject of this paper. The two effects compound.

The opportunity extends beyond established researchers. Someone outside a major university, someone returning after twenty years in industry, or a student without access to the right specialist now has a patient mathematical interlocutor. Its mistakes matter and its explanations

require scrutiny. But the prospect of serious mathematics becoming accessible to far more people is the reach channel of Section 5 applied to the producers of mathematics rather than its users.

The declaration describes the long journey from a difficult proof to a clear textbook explanation. Why assume the machine's contribution will be concentrated at the first end of that journey? Finding illuminating special cases, testing which hypotheses are necessary, suggesting alternative proofs, and translating between mathematical languages are all worthwhile targets, and they are exactly the activities that the historical record shows were undersupplied.

The chain the declaration describes is internal to mathematics, from proof to textbook, and the 68-year figure measures something external, from field to application. The two are in series: a result that has not yet reached a textbook has rarely reached an engineer. It was never the mathematics that took a lifetime to become useful. It was the translation, at both stages.

Read uncharitably, the declaration almost sounds as if its signatories would prefer their results not be used until much later, the way a studio prefers its film not go straight to a streaming service. That is surely not the intent, but the chain it defends has that effect. A single phrase can unlock immediate value. A hint such as *can the Reynolds projection simplify pandemic modelling* should not trickle through the system over decades if it can help now, and a model that has read both literatures can offer the hint the same afternoon.

There is also a quieter assumption in the declaration, that the people welcome in mathematics are those with the technique and the patience to turn insight into theorem. Technology moves on. There was a time when detailed knowledge of registers and buses was necessary to get useful work from a computer, and nobody now thinks the programmers of that era were the only real ones. Democratizing the proof may at first appear a threat. For some of us it is liberating: it removes the drudgery of turning insight into rigor, and leaves the insight. Rigor sometimes discovers that the insight was wrong, and a machine that supplies the rigor supplies that discovery too.

The institutional problems are real. Careers built around priority will need adjustment. Attribution must be preserved. Students need opportunities to develop independent judgment. Universities should reward the people who turn difficult results into usable understanding. Those are concrete tasks around which to organise an optimistic programme, and they are cheaper to carry out than leaving the transmission as slow as it has been.

Pure mathematics is a language, and a language should not be scarce. Nobody rations grammar. Yet the exchange of mathematical arguments between people who are not professional mathematicians remains rare, and the exchange between professionals moves at the pace of academic publishing, which is glacial. Perhaps one day it will be the norm in online forums to throw proofs and models back and forth the way code is thrown back and forth now, with a model doing the translation on each side.

This paper is an example. It was largely generated by a language model, in order to sharpen a point that began as a vague one: a class of six felt too small, and on reflection it looks absurd. There is tremendous value in refining vague arguments, and the refining used to cost a collaborator's month. The reader is welcome to point Claude, or any other model, at the repository in which this paper lives, github.com/microprediction/pure, provide scathing feedback, and improve the result.

The strongest response to a narrow benchmark culture is to demonstrate a richer mathematical practice: take a machine-assisted result and show how much understanding, teaching, and further discovery grow from it. Mathematicians have spent their lives wishing for more time, more collaborators, and better access to ideas. They may be getting some of those things. Surely that deserves at least a whistle.

11 Closing

We leave to future work the measurement of the reach multiplier, which is the assumption Table 1 depends on most. Two proxies suggest themselves: the share of model-assisted code changes that invoke a named mathematical result, and the lag between a mathematics preprint and its first appearance in an applied repository. Both are observable now, and neither was observable before.

There were six of us. That was too few on the record as it already stood, before language models: every named dismissal of a pure field in the survey has failed, and the underinvestment argument is sixty years old. There should have been a hundred.

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